

JAVA

Assignment-02

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classes using extends and implements:

1. C1 extends C2 (C-class)

Steps for execution:-

1. create a folder named assignment.

2. create a java file inside it:

C1 extends C2.java

3. write / save the program in the file.

4. Open the eclipse IDE.

5. compile the program:

6. Run the program:

7. Output:

This is class 2.

This is class 1.

2. C1 extends C2, C3:- (C-class)

Steps for execution:-

1. create a folder named assignment.

2. create a java file inside it:

C1 extends C2, C3.java

3. write / save the program in the file.

4. Open the eclipse IDE.

5. compile the program.

6. Run the program.

7. The code as currently written in can be compiled, because the invalid C1 declaration is commented out.

8. However, it cannot be run as a Java application because there is no `main()` method.

9. The compiler will show an error.

Note:- There is no output here, because C1 does not extend C2, C3. Java doesn't support multiple inheritance using classes.

3. C1 Implements I1.

Steps for execution:-

1. Create a folder named assignment.

2. Create a Java file inside it:

C1 implements I1.java.

3. Write / save the program in the file.

4. Open the Eclipse IDE.

5. Compile the program.

6. Run the program.

7. Output:

This is I1 interface.

This is C1 class.

4. C1 Implements I1, I2. (C - class, I - Interface)

Steps for execution:-

1. Open Eclipse IDE.

2. Create a Java project.

3. Create a package named oide.

4. Create a class named C1ImpI1AndI2.

5. Paste the code.

6. Save the program using ctrl+s.

7. Right-click the program.

8. Select Run as → Java Application

9. The main() method starts execution.

10. An object "obj" is created.

11. obj.sample() is executed.

12. Then obj.sum() is executed.

13. The output is displayed in the console.

14. Output:

This is in I1.

The sum of the numbers is 120.

This is in I2.

The sub of two numbers is: 80.

5. C1 implements C2 extends I1. (C-class, I-Interface)

Steps for execution:-

It is invalid.

1. Start.

2. Create class C2.

3. Create interface I1.

4. Try to create class C1.

5. Write C1 implements C2 extends I1.

6. Compile the program.

7. The compiler shows an error because the syntax is invalid.

8. Stop.

6. I_1 extends I_2 .

Steps for execution:-

1. create a folder named assignment.

2. create a Java file inside it:

I_1 extends I_2 .java.

3. write/save the program in the file.

4. open the eclipse IDE.

5. compile the program.

6. Run the program.

7. output:

I_2 method

I_1 method

7. I_1 implements C_1

Steps for execution:-

Note:- Here, I_1 is an interface and C_1 is a class. An interface can't

implement a class, so this is invalid.

1. create a folder named assignment.

2. create a Java file inside it:

I_1 extends I_2 .java.

3. write/save the program in the file.

4. open the eclipse IDE.

5. compile the program.

6. Run the program.

7. Try to compile/run the program.

8. The program itself has no main() method, so it cannot be executed as a Java application.

9. More importantly, if you actually write interface I_1 implements C_1 , the compiler will give an error.

8. I_1 extends I_2, I_3 (I-Interface)

Steps for execution:-

1. Create a folder named assignment.

2. Create a Java file inside it:

I_1 extends I_2 and I_3 .java

3. Write/Save the program in the file.

4. Open the Eclipse IDE.

5. Compile the program.

6. Run the program.

7. Output:

I_1 method

I_2 method

I_3 method